

Is robotics about to have its own ChatGPT moment?

1. **What is the problem that article (make sure to provide reference to base article) is highlighting and has been solved or needs to be solved? You should frame the key problem being discussed in the article you read as a single question or statement.**

Ans. There's a well-known observation among roboticists: What is hard for humans is easy for machines, and what is easy for humans is hard for machines. Called Moravec's paradox [1].

It is due to 3 main reasons:-

1. Robots lack precise control and coordination.
2. Their understanding of the surrounding world is limited because they are reliant on cameras and sensors to perceive it.
3. They lack an innate sense of practical physics.

2. **What would happen if this problem is solved? What is its impact?**

Ans. If scientists are able to connect AI with robotics they will be able to not only save a lot of money but also their time and energy and with the use of reinforced learning scientists will no longer have the need to waste their years in creating human centric designs.

At the Robotics Institute at Carnegie Mellon, where Pathak works, he says, "There used to be a saying that if you touch a robot, you add one year to your PhD." Now, he says, students get exposure to many robots and see results in a matter of weeks. [2]

In addition to that, if AI becomes a part of robotic system it will become more human friendly allowing everyone have advanced and easier access to these devices, making our lives easier in day to day tasks. Many roboticists think that large vision-language models, which are able to analyze image and language data, might offer robots important hints as to how the surrounding world works. They offer semantic clues about the world and could help robots with reasoning, deducing things, and learning by interpreting images [3].

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3. Since when has the world been working on this problem and what have we done?

Research the history of this problem and give a summarized account of happenings around this problem. Make sure to research how the previous solutions worked and why the previous solutions were not accepted as good solutions.

Ans. Despite the fast pace of development, robots still face many challenges before they can be released into the real world. They are still way too clumsy for regular consumers to justify spending tens of thousands of dollars on them. According to the text the issue is still not resolved as roboticist claims that they can potentially solve the problem but they cannot figure out how they will do that.

Why previous solutions failed:-

1. **Lack of adaptability and efficiency:** Robots were able to follow specific instructions and perform tasks but they were unable to adapt or understand it. Secondly, they were heavily programmed for that.
2. **Safety Concerns:** As AI in robotics began dealing with more complex tasks, unpredictability raised safety issues, particularly in applications like autonomous driving.

During 1950-1957, AI was a concept which was tested on different soft wares. It started when Japanese scientists thought of making an artificial mind which can think and learn just like humans. AI cherished maturity around 1962, after several tests and upgrades. The concept of artificial intelligence (AI) dates back several decades and has roots in the origins of modern computing. The history of AI includes the many ways television shows and movies use it as a simplistic catchall for the fear that technology is taking over our lives. *"The 1970s showed similar improvements, such as the first anthropomorphic robot being built in Japan, to the first example of an autonomous vehicle being built by an engineering grad student. However, it was also a time of struggle for AI research, as the U.S. government showed little interest in continuing to fund AI research."* [5]

In early 2000s several types of robots are built by different companies such as HONDA's ASIMO or MIT's KISMET. IBM's Watson natural language processing computer defeats two former champions on the television show "Jeopardy!" The number of internet users worldwide surpasses 4 billion.

4. What is the current approach as highlighted by article for solving this problem? Explain the technical details. You may not every technical detail pertinent to the problem, but the idea here is to capture system-level understanding and relate it to your previous knowledge. Recall the audience when you're writing your response, i.e. assume that the audience has the same background as your peers and address them.

Ans. To get better results with as less investment as possible scientists have come up with the idea of training robots through AI. Robots use cameras and sensors for control and coordination, roboticists plan the motions of their machines by writing a very specific list of actions for them to take. Robots trained like this are strictly choreographed to work in one specific setting. But how can we get software to learn multiple tasks and performing them for humans without much traditional planning and training.

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Roboticians have started using deep learning and neural networks to create systems that learn from their environment on the go and adjust their behavior. There are popular ways that researchers are using to train robots through AI, The technique mentioned in the respected article is reinforced learning which implies that systems improve through trial and error. Researchers idealized how human brains work using the aspect of memory they are now programming robots learn their movements, this helps them to teach themselves and learn new skills on the go.

Another popular technique used by roboticians is imitation learning in which they learn new skills by imitating humans through their cameras and sensors. By using these techniques researchers claim to have taught robots more than 200 skills, such as peeling vegetables and pouring liquids, and say they are working toward teaching 1,000 skills by the end of the year.

An example (Robots meet AI, paragraph 10) published in the following articles explains how a team managed to save \$32,000 by using AI. They taught their machine the very basic task from tearing a paper towel to cooking shrimps. They claim that AI can help robots acquire transferable skills: training on one task can improve its performance for others.

Reinforcement learning (RL) is another key technique used to make robots more adaptive in dynamic environments. In RL, an AI agent learns through trial and error by receiving rewards or penalties based on the actions it takes. Over time, the robot optimizes its behavior to achieve the best outcomes. [4]

5. What further work needs to be done on this problem?

Ans. While scientists are doing great progress in making the most out of AI tools to help them in robotics there is always space for betterment. As explained in the article robots are not too efficient to fulfill human requirements, robots lack the capability of multitasking which can consume a lot of time, they are a little clumsy a few times which can cause humans to face loss, they also lack common sense calculating everything logically. As stated in the article (under the heading of missing pieces, paragraph 1 "What is hard for humans is easy for machines, and what is easy for humans is hard for machines") this line clearly explains how robots can perform complex tasks but fail in the basic tasks. To sum up, scientists need to work on the efficiency of their AI powered robots but also teach them simple tasks.

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REFERENCES:-

[1] MIT Technology Review: Is robotics about to have its own ChatGPT moment? (Topic: Missing pieces)

Link: <https://technologyreview.com/2024/04/11/1090718/household-robots-ai-data-robotics/>

[2] MIT Technology Review: Is robotics about to have its own ChatGPT moment? (Topic: Robots meet AI)

[3] MIT Technology Review: Is robotics about to have its own ChatGPT moment? (Topic: More is more, related stories)

[4] Source: OpenAI; ChatGPT (prompt: What is the current approach by roboticist for solving problems faced in past while integrating robots with Artificial Intelligence.)

[5] Tableau.com

Link: <https://www.tableau.com/data-insights/ai/history#:~:text=1929%3A%20Japanese%20professor%20Makoto%20Nishimura,of%20computers%20to%20human%20brains.>